

Lecture 11

Inherited Properties of Metric Spaces

- $d_x(p, q)$, by definition, is the same as $d_{M(p,q)}$
- A set U is open in M iff $U \cap X$ is open with respect to d_X
 - $M_r(p) \cap X$ is the r -neighborhood of p using d_X , alternatively written as $X_r(p)$
 - $M_r(p) \cap X \subset U \cap X$
 - Topology in M “reduces” through intersection
- Similar idea for closed sets
 - For a closed set $K \subset X$, $\exists L \subset M$ s.t. $L \cap X = K$
 - L is closed in M
 - Using sequences to show

Antoine Necklaces

- Solid torus is $D^2 \times S^1$, torus shell is $S^1 \times S^1$
- Neighborhoods in $\mathbb{R}^3$ can be weird shapes like tori
- Theorem statement: $\exists A \subset \mathbb{R}^3$, $A \cong C$ s.t. $\exists$ loop in $\mathbb{R}^3$ not contractable in $\mathbb{R}^3 \setminus A$
- A embedded of $\mathbb{R}^3 \cong C$ embedded in $\mathbb{R}^3$
 - No ambient homeomorphism
- Using the idea of making each torus a necklace
 - A_1 is turning a torus into a necklace of tori
 - A_2 iterates the step performed in A_1
 - Can keep repeating to A_n and beyond
- $A = \bigcap_{n=1}^{\infty} A_n$
- For $p \in A$, there will be some small solid torus in A_n that contains p
 - It's not totally disconnected, but pretty disconnected
 - Can keep iterating
- Clopen sets are inherited
- A is totally disconnected
 - p is in some small torus, and we have some q_n in some torus of A_n that is nearby
 - p will never be q_n

Homotopy

- Function $\mathcal{H}(t, \theta)$, $t \in [0, 1]$, $\theta \in [0, 2\pi]$
 - Circle governed by time with soem continuous movement
- Let's say we have some Antoine necklace looping around the z -axis
 - We form some circle by connected ends of a segment on the z -axis
- Not contractable means that this circle of the z -axis cannot be shrunk to the origin without hitting the necklace
- More formally, the statement looks something like $\nexists$ homotopy $\mathcal{H}(0, \theta) =$ original loop of the z -axis and $\mathcal{H}(1, \theta) =$ origin and $\mathcal{H}([0, 1] \times S^1) \cap A = \emptyset$
 - The last condition means the movement doesn't hit the Antoine necklace
- Hand-wavy idea: think of necklace as loop, can see how it's 2 linked circles

Constructing $\mathbb{R}$ from $\mathbb{Q}$ using Cauchy Sequences

- Cocaluchy equivalence relation $(x_n) \sim (y_n)$ on M, d
 - $d(x_n, y_n) \rightarrow 0$ as $n \rightarrow \infty$
- Can create equivalence classes $[(x_n)] = \{(y_n) : (y_n) \sim (x_n)\}$
- Cuts define topology of $\mathbb{R}$, want to use Cauchy sequences for algebraic operations

- Use sequences in $\mathbb{Q}$, with $d = |x - y|$ for $x, y \in \mathbb{R}$
- $\mathbb{E}$ = set of Cofinite equivalence classes in $\mathbb{Q}$
- Need to show $\mathbb{E}$ is a field
- Addition
 - Defined naturally, $[(x_n)] + [(y_n)] = [(x_n + y_n)]$
 - $0^* = [(0, 0, 0, \dots)]$
- Multiplication
 - $[(x_n)] \cdot [(y_n)] = [(x_n \cdot y_n)]$
 - A bit harder to show sequence is still Cauchy, but it is doable
 - $1^* = [(1, 1, 1, \dots)]$
- $\mathbb{Q}$ is a field, so $\mathbb{E}$ is a field
- Create a bijection $L : \mathbb{E} \rightarrow \mathbb{R}$, $\mathbb{R}$ is the real numbers constructed from cuts
- $\mathbb{Q} \subset \mathbb{E}$
- $L : [(x_n)] \rightarrow \lim_{n \rightarrow \infty} x_n$ in $\mathbb{R}$
 - $\mathbb{E} \rightarrow \mathbb{R}, \mathbb{E}_{\mathbb{Q}} \rightarrow \mathbb{Q}$
- L is a bijection, also a field isomorphism
- Essentially “force” field nature onto $\mathbb{R}$